Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology**	
Subject: DSIP	**Aim: Simulate discrete time sequences.**	
Experiment No:1	**Date:15-07-2026**	**Enrollment No:92400133063**

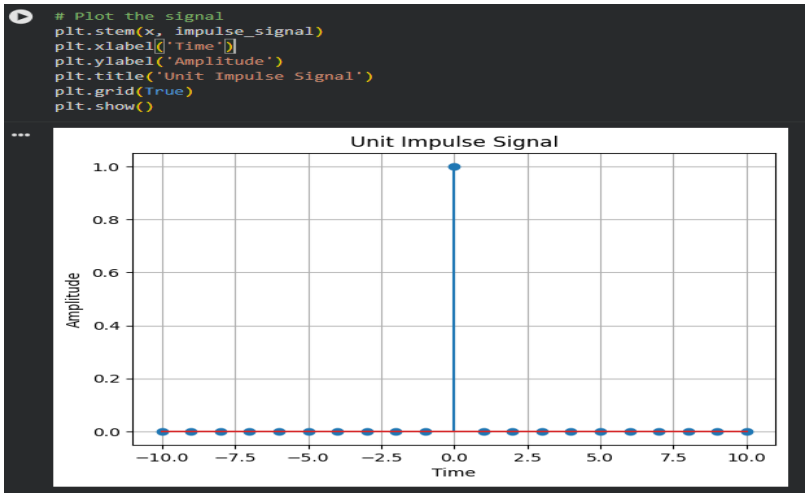
Experiment:-1

AIM:- Simulate Discrete time Sequences

Code:-

```
import numpy as np
import matplotlib.pyplot as plt
def unit_impulse(length, position):
    signal = np.zeros(length)
    signal[position] = 1
    return signal
# Parameters
start = -10 # Start value of the x-axis range
stop = 10 # Stop value of the x-axis range
step = 1 # Step size
# Generate x-axis values
x = np.arange(start, stop+step, step)
# Generate unit impulse signal
impulse_signal = unit_impulse(len(x), abs(start)//step)
# Plot the signal
plt.stem(x, impulse_signal)
plt.xlabel('Time')
plt.ylabel('Amplitude')
plt.title('Unit Impulse Signal')
plt.grid(True)
plt.show()
```

Output:-



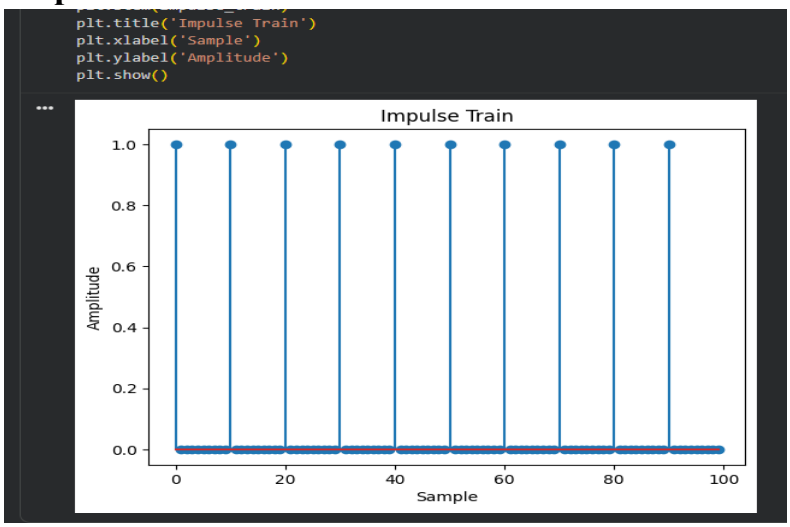
Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology**	
Subject: DSIP	**Aim: Simulate discrete time sequences.**	
Experiment No:1	**Date:15-07-2026**	**Enrollment No:92400133063**

AIM:- Simulate impulse train

Code:-

```
import numpy as np
import matplotlib.pyplot as plt
def simulate_impulse_train(signal_length, period):
 impulse_train = np.zeros(signal_length)
 for n in range(signal_length):
 if n % period == 0:
 impulse_train[n] = 1
 return impulse_train
# Define the parameters for the impulse train
signal_length = 100 # Length of the impulse train
period = 10 # Period of the impulse train
# Simulate the impulse train
impulse_train = simulate_impulse_train(signal_length, period)
# Plot and display the impulse train
plt.stem(impulse_train)
plt.title('Impulse Train')
plt.xlabel('Sample')
plt.ylabel('Amplitude')
plt.show()
```

output:-



Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology**	
Subject: DSIP	**Aim: Simulate discrete time sequences.**	
Experiment No:1	**Date:15-07-2026**	**Enrollment No:92400133063**

AIM:Write a python program to Simulate continuous and discrete unit step signal.

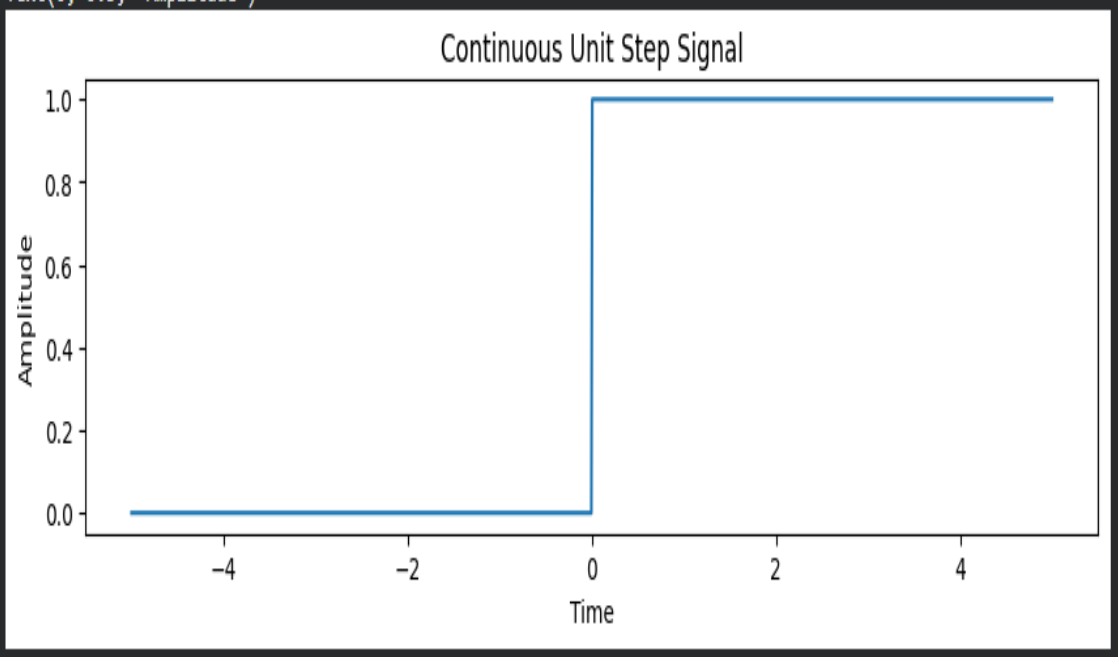
code:-

```
import numpy as np
import matplotlib.pyplot as plt
def simulate_continuous_exponential(time, amplitude, coefficient):
 exponential_signal = amplitude * np.exp(coefficient * time)
 return exponential_signal
def simulate_discrete_exponential(num_samples, amplitude, coefficient):  exponential_signal =
amplitude * np.exp(coefficient * np.arange(num_samples))  return exponential_signal
# Define the time range for the continuous exponential signal
time = np.linspace(0, 5, 1000) # Time range from 0 to 5
# Define the number of samples, initial amplitude, and coefficient for the discrete
exponential  signal
num_samples = 20 # Number of samples
amplitude = 2 # Initial amplitude
coefficient = -0.5 # Exponential coefficient
# Simulate the continuous exponential signal
continuous_exponential = simulate_continuous_exponential(time, amplitude, coefficient)
# Simulate the discrete exponential signal
discrete_exponential = simulate_discrete_exponential(num_samples, amplitude, coefficient)
# Plot and display the continuous and discrete exponential signals
plt.figure(figsize=(10, 6))
plt.subplot(2, 1, 1)
plt.plot(time, continuous_exponential)
plt.title('Continuous Exponential Signal')
plt.xlabel('Time')
plt.ylabel('Amplitude')
plt.subplot(2, 1, 2)
plt.stem(discrete_exponential)
plt.title('Discrete Exponential Signal')
plt.xlabel('Sample')
plt.ylabel('Amplitude')
plt.tight_layout()
plt.show()
```

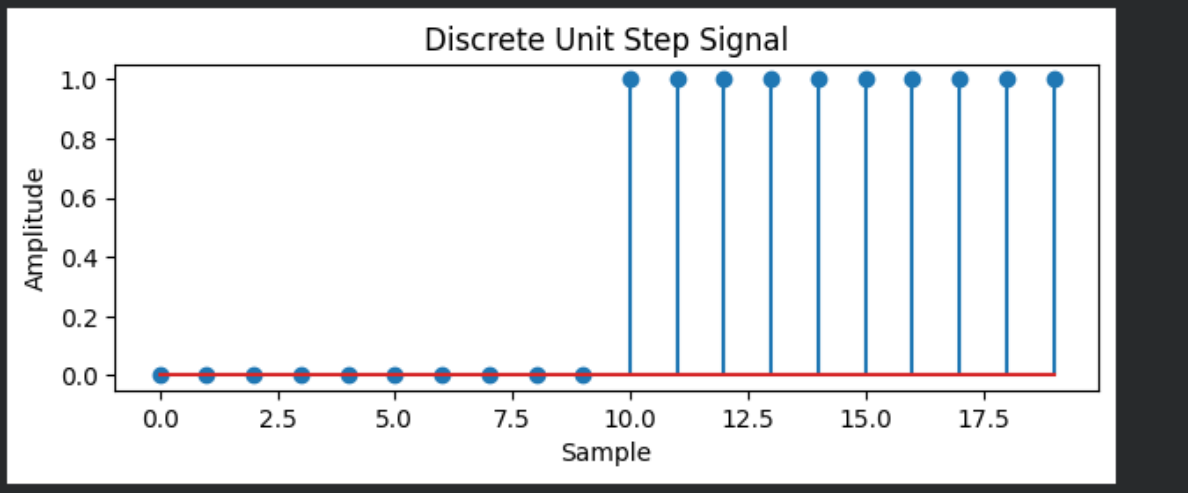
Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology**	
Subject: DSIP	**Aim: Simulate discrete time sequences.**	
Experiment No:1	**Date:15-07-2026**	**Enrollment No:92400133063**

output:-

```
Text(0, 0.5, 'Amplitude')
```




```
# np.savetxt('discrete_unit_step.txt', discrete_unit_step, delimiter=',')
```



Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology**	
Subject: DSIP	**Aim: Simulate discrete time sequences.**	
Experiment No:1	**Date:15-07-2026**	**Enrollment No:92400133063**

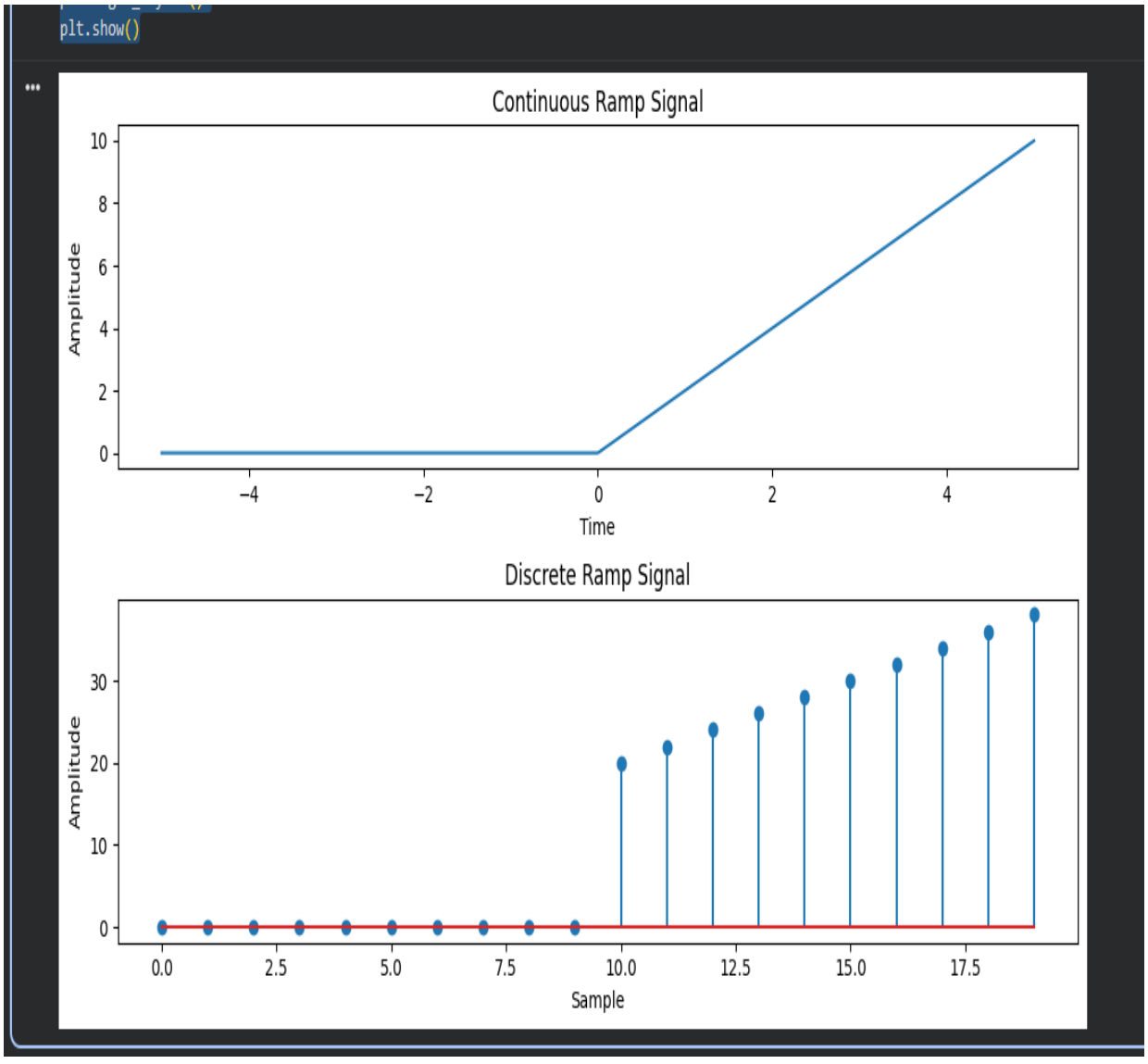
AIM:- Write a python program to Simulate continuous and discrete ramp signal.

Code:-

```
import numpy as np
import matplotlib.pyplot as plt
def simulate_continuous_ramp(time, slope):
  ramp = np.zeros_like(time)
  ramp[time >= 0] = slope * time[time >= 0]
  return ramp
def simulate_discrete_ramp(num_samples, slope):
 ramp = np.zeros(num_samples)
 ramp[num_samples // 2:] = slope * np.arange(num_samples // 2, num_samples)
 return ramp
# Define the time range for the continuous ramp signal
time = np.linspace(-5, 5, 1000) # Time range from -5 to 5
# Define the number of samples and slope for the discrete ramp signal
num_samples = 20 # Number of samples
slope = 2 # Slope of the ramp
# Simulate the continuous ramp signal
continuous_ramp = simulate_continuous_ramp(time, slope)
# Simulate the discrete ramp signal
discrete_ramp = simulate_discrete_ramp(num_samples, slope)
# Plot and display the continuous and discrete ramp signals
plt.figure(figsize=(10, 6))
plt.subplot(2, 1, 1)
plt.plot(time, continuous_ramp)
plt.title('Continuous Ramp Signal')
plt.xlabel('Time')
plt.ylabel('Amplitude')
plt.subplot(2, 1, 2)
plt.stem(discrete_ramp)
plt.title('Discrete Ramp Signal')
plt.xlabel('Sample')
plt.ylabel('Amplitude')
plt.tight_layout()
plt.show()
```

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology**	
Subject: DSIP	**Aim: Simulate discrete time sequences.**	
Experiment No:1	**Date:15-07-2026**	**Enrollment No:92400133063**

output:-



Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology**	
Subject: DSIP	**Aim: Simulate discrete time sequences.**	
Experiment No:1	**Date:15-07-2026**	**Enrollment No:92400133063**

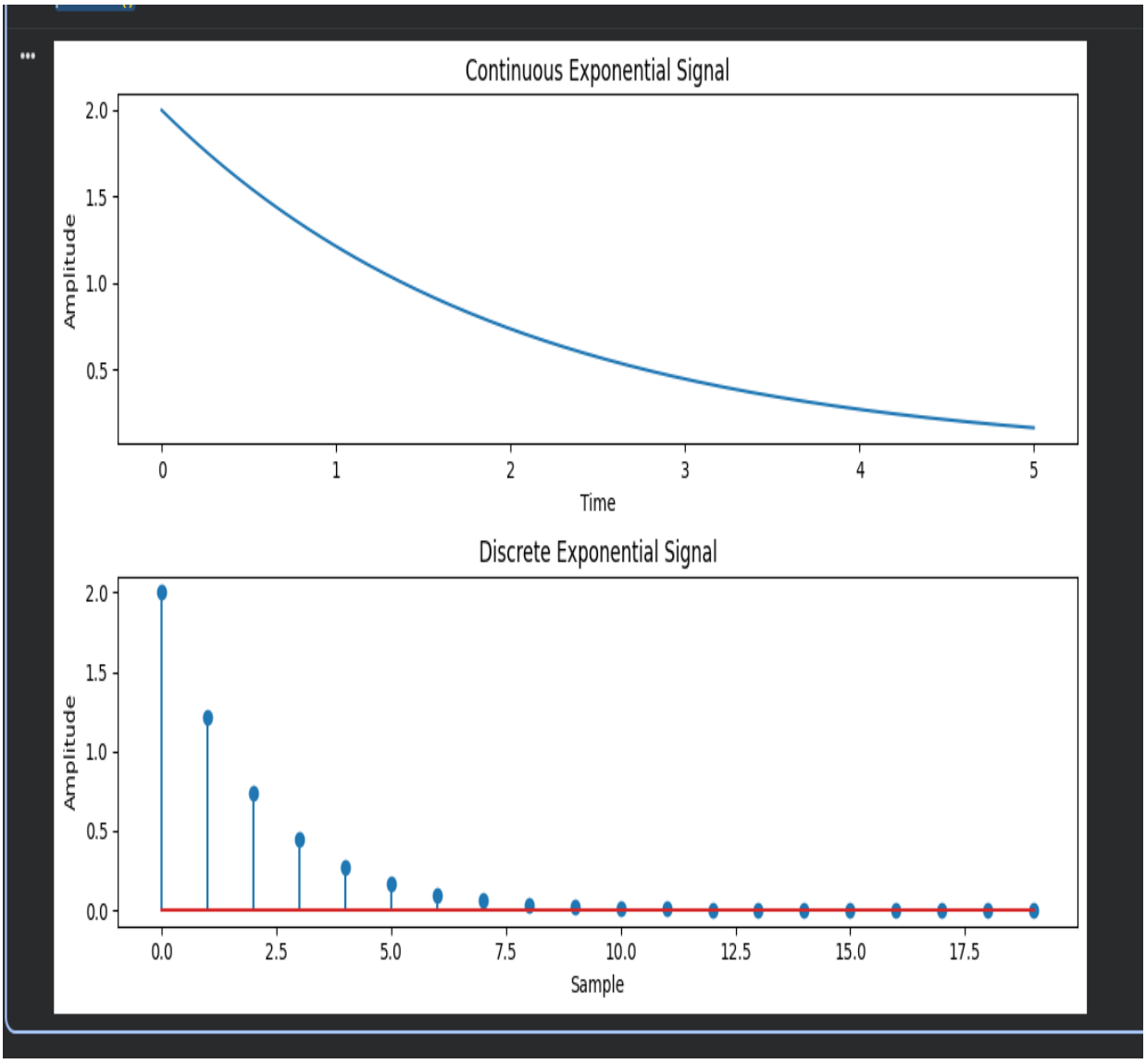
AIM:- Write a python program to Simulate continuous and discrete exponential signal.

Code:-

```
def simulate_continuous_exponential(time, amplitude, coefficient):
 exponential_signal = amplitude * np.exp(coefficient * time)
 return exponential_signal
def simulate_discrete_exponential(num_samples, amplitude, coefficient):
 exponential_signal = amplitude * np.exp(coefficient * np.arange(num_samples))
 return exponential_signal
# Define the time range for the continuous exponential signal
time = np.linspace(0, 5, 1000) # Time range from 0 to 5
# Define the number of samples, initial amplitude, and coefficient for the discrete exponential
# signal
num_samples = 20 # Number of samples
amplitude = 2 # Initial amplitude
coefficient = -0.5 # Exponential coefficient
# Simulate the continuous exponential signal
continuous_exponential = simulate_continuous_exponential(time, amplitude, coefficient)
# Simulate the discrete exponential signal
discrete_exponential = simulate_discrete_exponential(num_samples, amplitude, coefficient)
# Plot and display the continuous and discrete exponential signals
plt.figure(figsize=(10, 6))
plt.subplot(2, 1, 1)
plt.plot(time, continuous_exponential)
plt.title('Continuous Exponential Signal')
plt.xlabel('Time')
plt.ylabel('Amplitude')
plt.subplot(2, 1, 2)
plt.stem(discrete_exponential)
plt.title('Discrete Exponential Signal')
plt.xlabel('Sample')
plt.ylabel('Amplitude')
plt.tight_layout()
plt.show()
```

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology**	
Subject: DSIP	**Aim: Simulate discrete time sequences.**	
Experiment No:1	**Date:15-07-2026**	**Enrollment No:92400133063**

output:-



Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology**	
Subject: DSIP	**Aim: Simulate discrete time sequences.**	
Experiment No:1	**Date:15-07-2026**	**Enrollment No:92400133063**

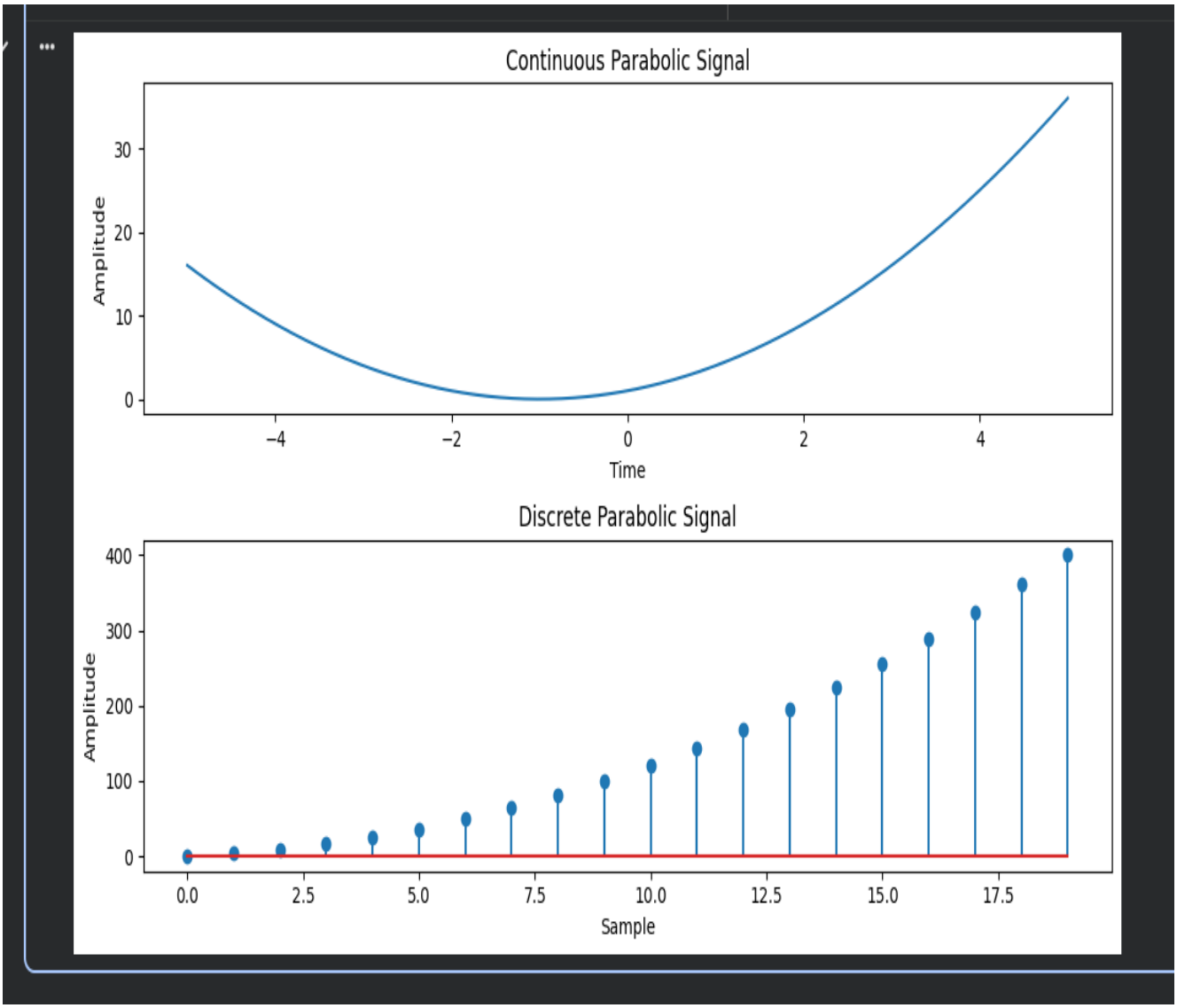
AIM:- Write a python program to Simulate continuous and discrete parabolic signal.

Code:-

```
def continuous_parabolic(time, coefficients):
 parabolic_signal = np.polyval(coefficients, time)
 return parabolic_signal
def simulate_discrete_parabolic(num_samples, coefficients):
 parabolic_signal = np.polyval(coefficients, np.arange(num_samples))
 return parabolic_signal
# Define the time range for the continuous parabolic signal
time = np.linspace(-5, 5, 1000) # Time range from -5 to 5
# Define the number of samples and coefficients for the discrete parabolic signal
num_samples = 20 # Number of samples
coefficients = [1, 2, 1] # Coefficients of the parabolic signal
# Simulate the continuous parabolic signal
continuous_parabolic = continuous_parabolic(time, coefficients)
# Simulate the discrete parabolic signal
discrete_parabolic = simulate_discrete_parabolic(num_samples, coefficients)
# Plot and display the continuous and discrete parabolic signals
plt.figure(figsize=(10, 6))
plt.subplot(2, 1, 1)
plt.plot(time, continuous_parabolic)
plt.title('Continuous Parabolic Signal')
plt.xlabel('Time')
plt.ylabel('Amplitude')
plt.subplot(2, 1, 2)
plt.stem(discrete_parabolic)
plt.title('Discrete Parabolic Signal')
plt.xlabel('Sample')
plt.ylabel('Amplitude')
plt.tight_layout()
plt.show()
```

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology**	
Subject: DSIP	**Aim: Simulate discrete time sequences.**	
Experiment No:1	**Date:15-07-2026**	**Enrollment No:92400133063**

output:-



Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology**	
Subject: DSIP	**Aim: Simulate discrete time sequences.**	
Experiment No:1	**Date:15-07-2026**	**Enrollment No:92400133063**

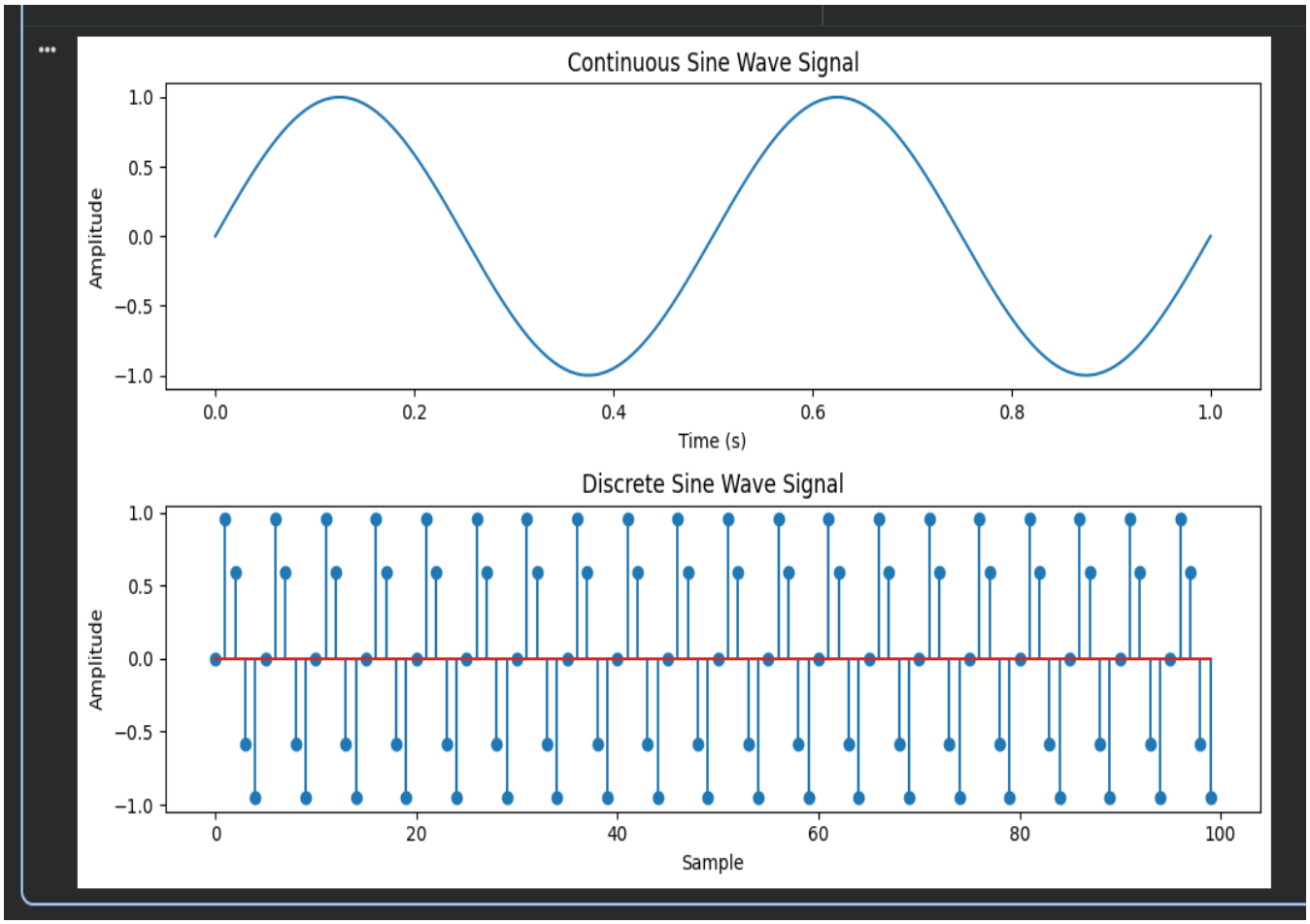
AIM:- Write a python program to Simulate continuous and discrete sine wave signal.

Code:-

```
def simulate_continuous_sine_wave(time, amplitude, frequency, phase):
 sine_wave = amplitude * np.sin(2 * np.pi * frequency * time + phase)
 return sine_wave
def simulate_discrete_sine_wave(num_samples, sampling_frequency, amplitude, frequency,
phase):
 time = np.arange(num_samples) / sampling_frequency
 sine_wave = amplitude * np.sin(2 * np.pi * frequency * time + phase)
 return sine_wave
# Define the time range for the continuous sine wave signal
time = np.linspace(0, 1, 1000) # Time range from 0 to 1 second
# Define the number of samples, sampling frequency, and parameters for the discrete sine wave
signal
num_samples = 100 # Number of samples
sampling_frequency = 10 # Sampling frequency in Hz
amplitude = 1 # Amplitude of the sine wave
frequency = 2 # Frequency of the sine wave in Hz
phase = 0 # Phase angle of the sine wave in radians
# Simulate the continuous sine wave signal
continuous_sine_wave = simulate_continuous_sine_wave(time, amplitude, frequency, phase)
# Simulate the discrete sine wave signal
discrete_sine_wave = simulate_discrete_sine_wave(num_samples, sampling_frequency,
amplitude, frequency, phase)
# Plot and display the continuous and discrete sine wave signals
plt.figure(figsize=(10, 6))
plt.subplot(2, 1, 1)
plt.plot(time, continuous_sine_wave)
plt.title('Continuous Sine Wave Signal')
plt.xlabel('Time (s)')
plt.ylabel('Amplitude')
plt.subplot(2, 1, 2)
plt.stem(discrete_sine_wave)
plt.title('Discrete Sine Wave Signal')
plt.xlabel('Sample')
plt.ylabel('Amplitude')
plt.tight_layout()
plt.show()
```

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology**	
Subject: DSIP	**Aim: Simulate discrete time sequences.**	
Experiment No:1	**Date:15-07-2026**	**Enrollment No:92400133063**

output:-



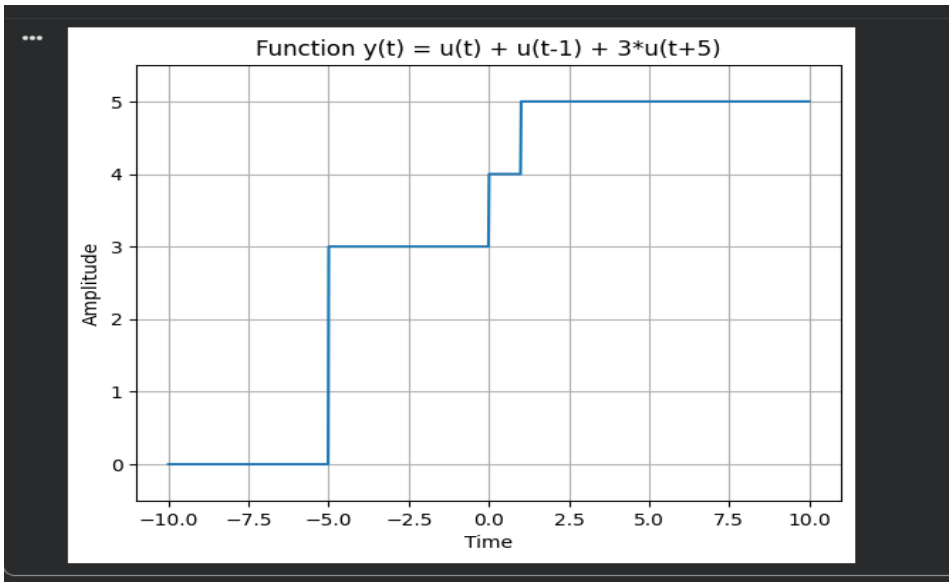
Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology**	
Subject: DSIP	**Aim: Simulate discrete time sequences.**	
Experiment No:1	**Date:15-07-2026**	**Enrollment No:92400133063**

AIM:- Write a python program to simulate y(t)=u(t)+u(t-1)+3u(t+5).

Code:-

```
def simulate_function(time):
 y = np.zeros_like(time)
 y[time >= 0] = 1
 y[time >= 1] += 1
 y[time >= -5] += 3
 return y
# Define the time range
time = np.linspace(-10, 10, 1000)
# Simulate the function
function_values = simulate_function(time)
# Plot and display the function
plt.plot(time, function_values)
plt.title('Function y(t) = u(t) + u(t-1) + 3*u(t+5)')
plt.xlabel('Time')
plt.ylabel('Amplitude')
plt.ylim([-0.5, 5.5])
plt.grid(True)
plt.show()
```

output:-



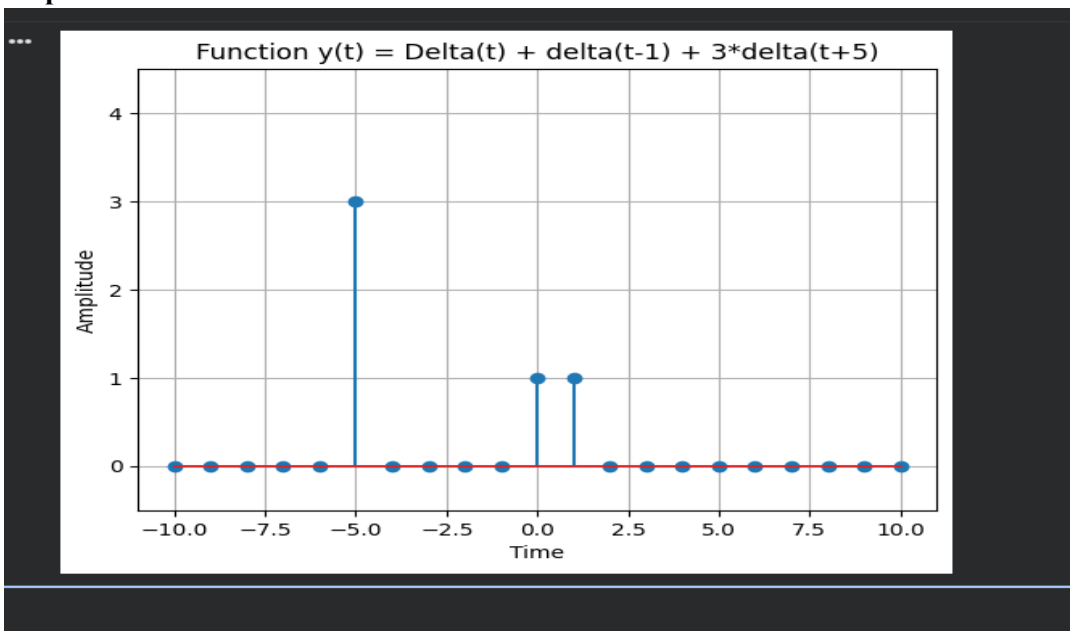
Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology**	
Subject: DSIP	**Aim: Simulate discrete time sequences.**	
Experiment No:1	**Date:15-07-2026**	**Enrollment No:92400133063**

AIM:-Write a python program to simulate y(t)=Delta(t)+delta(t-1)+3*delta(t+5).

Code:-

```
def simulate_function(time):
 y = np.zeros_like(time)
 y[time == 0] = 1
 y[time == 1] += 1
 y[time == -5] += 3
 return y
# Define the time range
time = np.arange(-10, 11)
# Simulate the function
function_values = simulate_function(time)
# Plot and display the function
plt.stem(time, function_values)
plt.title('Function y(t) = Delta(t) + delta(t-1) + 3*delta(t+5)')
plt.xlabel('Time')
plt.ylabel('Amplitude')
plt.ylim([-0.5, 4.5])
plt.grid(True)
plt.show()
```

output:-



Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology**	
Subject: DSIP	**Aim: Simulate discrete time sequences.**	
Experiment No:1	**Date:15-07-2026**	**Enrollment No:92400133063**

Conclusion:-

Signal Type	Continuous Form	Discrete Form	Key Parameter
Unit Impulse	$\delta(t)$	$\delta[n]$	Position at n=0
Impulse Train	Periodic $\delta(t)$	Periodic $\delta[n]$	Period
Unit Step	$u(t)$	$u[n]$	Transition at t=0
Ramp	$r(t) = a \cdot t$	$r[n] = a \cdot n$	Slope (a)
Exponential	$A \cdot e^{(bt)}$	$A \cdot e^{(bn)}$	Coefficient (b)
Parabolic	at^2+bt+c	an^2+bn+c	Coefficients [a,b,c]
Sine Wave	$A \cdot \sin(2\pi ft+\varphi)$	$A \cdot \sin(2\pi fn/Fs+\varphi)$	Frequency (f)
Composite Signals	$u(t)+u(t-1)+3u(t+5)$	$\delta(t)+\delta(t-1)+3\delta(t+5)$	Multiple components

The modular function-based approach allows for easy modification of parameters and extension to more complex signals. This implementation provides a solid foundation for advanced topics in digital signal processing, communications, and control systems. The ability to simulate both continuous and discrete signals in the same framework enhances understanding of the relationship between analog and digital signal representations.